NYAYA AI

Consumer Rights Assistant
Solution Report

NAME	ROLL NUMBER
Harshita	**228170445**

GitHub Repository	https://github.com/harshita852/Nyaya-AI
Live Deployment	https://nyaya-ai-harshta.streamlit.app/

1. Problem Statement

Consumers often face difficulties when products are defective, refunds are delayed, services are not delivered, or sellers fail to respond. Many are unfamiliar with consumer-protection procedures and do not know what information to collect, which channel to approach, or how to write a clear complaint. Nyaya AI aims to make the first step easier by turning a grievance description into a structured, understandable action plan.

2. Proposed Solution

Nyaya AI is a web-based consumer-rights assistance prototype built for Indian consumers. A user describes a problem in natural language and selects a preferred response language. The application sends the description to Google Gemini with task-specific instructions. The model returns structured information that the application presents as a case overview, preliminary legal analysis, suggested next steps, evidence checklist, timeline, and editable complaint letter.

The tool provides preliminary guidance and document preparation. It does not replace a lawyer, a consumer commission, or an official grievance authority.

3. System Architecture and Technology Stack

The prototype uses a lightweight Python architecture. Streamlit provides the user interface and session state; the Google GenAI Python SDK communicates with the configured Gemini model; prompt instructions define the analysis task and expected JSON response; and python-docx creates a downloadable complaint document.

Frontend: Streamlit for language selection, complaint input, result tabs, and downloads.

Backend: Python modules for prompt construction, model requests, response parsing, and document generation.

AI service: Google Gemini through the `google-genai` SDK.

Configuration: Environment variables loaded from a local `.env` file; secrets are excluded from the repository.

Deployment: Streamlit Community Cloud.

4. Key Functionalities

4.1 Complaint intake and language selection

Users enter a free-form description and request a response in English, Hindi, or Hinglish. A sample complaint demonstrates the expected input.

4.2 Structured case analysis

The model extracts key facts, summarises the dispute, organises a chronology, identifies missing details, explains potentially relevant consumer rights, and suggests practical actions. The interface separates these outputs into readable tabs.

4.3 Evidence checklist and complaint drafting

The application highlights potentially useful evidence, such as invoices, order confirmations, payment records, screenshots, warranty documents, and seller communications. It also produces an editable complaint draft that can be downloaded as a Word document.

4.4 Structured output

The model is asked to return JSON with defined fields. The application parses the response and supplies defaults for missing fields so results can be displayed consistently.

5. Legal Reliability and Hallucination Prevention

The system prompt instructs the model not to invent statutory sections, judgments, citations, facts, or URLs. It asks the model to state uncertainty, identify missing information, avoid guaranteeing outcomes, and prefer official sources. The Consumer Protection Act, 2019 and the National Consumer Helpline are included as reference points where relevant.

Important limitation: The current prototype uses prompt-based generation. It does not perform retrieval-augmented generation (RAG), independently verify every legal statement, or guarantee that linked sources are current. Users should verify important information with official authorities or a qualified legal professional.

6. Testing and Evaluation

The solution was considered against representative scenarios: a defective product, a delayed refund, a paid service that was not delivered, an incomplete complaint, and a prompt containing a fabricated statutory reference. These cases exercise both the normal analysis flow and the instruction to request clarification or reject unsupported legal claims.

- Check that key facts and dates are preserved rather than altered.
- Check that missing details are requested instead of assumed.
- Check that suggested actions and evidence are relevant to the dispute.
- Check that complaint drafts remain editable and do not introduce unsupported facts.
- Check that fabricated legal provisions are not presented as genuine.

Evaluation should use actual outputs captured from the deployed application. Prepared sample outputs should be labelled illustrative, not represented as live model test results.

7. Limitations and Future Scope

The quality of the analysis depends on the completeness of the user's description and the model's response. The prototype does not submit complaints to authorities, provide case-specific legal representation, or maintain a verified legal knowledge base. Future improvements could include a curated and regularly updated legal-source repository, retrieval-based citations, stronger validation of URLs and statutory references, multilingual evaluation, user accounts, and guided complaint-submission workflows.

8. Conclusion

Nyaya AI demonstrates how a conversational AI interface can help consumers organise a grievance, understand possible next steps, identify supporting evidence, and prepare a first-draft complaint. By combining a simple Streamlit interface with structured Gemini responses and downloadable documents, the prototype provides an accessible starting point for consumer grievance preparation while communicating the need to verify legal information.